

ESP32-S3-Based Automated 180° Turntable for Jewelry Product Photography

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Abstract— The photography of jewelry products needs consistent positioning in order to get different angles without causing any disturbance to the product. In this project, an ESP32-S3-based automated 180° turntable has been developed for positioning of jewelry products. A 28BYJ-48 5 V geared stepper motor is driven through a ULN2003 motor-driver interface controlled by ESP32-S3 pins D2–D5. Motor speed is adjusted in software using a 10 k Ω potentiometer connected to A0. A momentary push-button connected to D6 provides local control. The motor is powered by an external 5 V, 2 A power source, and BLE connectivity is provided directly by the ESP32-S3 module. Approximately 2048 half-steps are used for nominal 180° rotation. Bench testing successfully verified bidirectional rotation, speed control, push-button operation, nominal 180° rotation, and BLE connectivity.

Keywords— ESP32-S3, Automated Turntable, Jewelry Photography, Stepper Motor, ULN2003 Motor Driver, 180° Rotation, Embedded Systems, Bluetooth Low Energy, Product Photography, Motor Speed Control

I. INTRODUCTION

Visual aspects play a huge role in online sales of any products including jewelry since customers do not have a chance to check the product visually in advance. Therefore, photographs of the product should convey not only the front view of the product but also its thickness, side view, shape, polishing, stone positions and so forth.

In particular, this is especially important for small objects like earrings, rings, necklaces and other lightweight jewelry products. Some of the aspects which might be seen in one position will be visible only with some shift of the view angle.

In general, the standard workflow for photographing usually includes manual rotation or moving of the product after each shot. In this case, the required equipment is minimal, but there might be operator-dependent variations. For example, in case the photographer wants to make a 30° turn of the object, actually, it might be some other degrees. Touching of the product during shooting may also affect its exact positioning.

The problem becomes more critical when a number of similar products is being photographed in a unified catalog form. Controlled platform may help to reduce the repeated manipulations with the product and also may ensure more accurate positioning.

Therefore, the objective of the present project is to design a portable automated turntable capable of rotating a lightweight jewelry product through an approximately 180° angular range. It is designed to be bidirectional. The turntable rotates to a predetermined point of 180° from the start and then continues its rotation in the opposite direction towards the start.

In the present project, the focus is made on the positioning automation of the product. The camera and lighting system are not automatically controlled by the ESP32-S3. Therefore, the contribution of the current project is mainly in the embedded electromechanical positioning system.

The developed prototype combines:

1. ESP32-S3 embedded control
2. Stepper motor sequencing
3. ULN2003 motor-driver interfacing
4. Potentiometer-based speed adjustment
5. Push-button control
6. Bluetooth Low Energy communication
7. External motor power
8. Compact mechanical integration

Key contributions from this study include the design of a bidirectional 180° jewelry-positioning mechanism, incorporation of local and wireless control capabilities, and the creation of a compact design where most electronic and mechanical parts are housed under the rotating base.

II. SYSTEM REQUIREMENTS AND DESIGN CONSIDERATIONS

A. 180° Operating range:

The principal functional requirement is an approximate 180° operating range rather than continuous 360° rotation.

The motion can be represented as:

$$0^\circ - 180^\circ$$

Followed by:

$$180^\circ - 0^\circ$$

It is possible to change the view of the jewelry in many different positions without having to make a full turn to get back to its original position.

The range of 180° operation is especially suitable for taking pictures of the product from different angles (front, side, intermediate and opposite side) while leaving the camera position unchanged.

B. Bidirectional Operation:

The motor should be capable of rotating in both directions. Reversal of the rotation direction is done purely in software by changing the sequence of switching on motor phases. Thus it becomes unnecessary to use an additional hardware circuit for controlling the direction of motor rotation.

C. Adjustable Rotational Speed:

Jewelry photography does not require high-speed rotation.

Slow and steady rotation is preferred since quick motion might move the product or cause vibrations. To allow adjusting the speed a 10 k Ω potentiometer is provided. Rather than changing the motor supply voltage, the potentiometer adjusts the delay between consecutive motor steps.

D. Local Control:

Local control is useful during bench testing, mechanical assembly and troubleshooting.

A physical momentary push-button allows the system to be operated without requiring wireless communication.

E. Bluetooth Capability:

Remote control was thought to be useful as touching the turntable after positioning the jewelry may have caused some disturbance to the jewelry placement.

The ESP32-S3 provides integrated Bluetooth Low Energy capability; therefore, no external Bluetooth module is required.

F. Electrical Requirements:

The motor cannot receive power through the GPIO pins of the ESP32-S3.

Only logic level signals from the microcontroller go to the ULN2003.

Power supply for the motor comes separately from a 5 V, 2 A power adapter.

G. Mechanical Packaging:

One of the important mechanical requirements is that the electronics should be housed under the platform.

At present, the housing offers about 9 cm vertical clearance. Thus, all the motor components need to fit in this space and not interfere with rotation.

III. HARDWARE AND ELECTRICAL IMPLEMENTATION

A. ESP32-S3 Microcontroller:

The ESP32-S3 was selected as the primary controller because of its compact size, GPIO availability, analog-input capability, and integrated wireless communications.

The controller performs the following functions:

1. Reads the analog potentiometer input.
2. Detects the push-button operation.
3. Generates the motor half-step sequence.
4. Controls motor direction.
5. Controls the number of motor steps.
6. Controls the speed of the motor.
7. Provides Bluetooth communication.
8. Supports future automated camera-control functionality.

B. Implemented Pin Assignment:

COMPONENT	ESP32-S3 CONNECTION	FUNCTION
ULN2003 IN1	D2	Motor Phase Control
ULN2003 IN2	D3	Motor Phase Control

ULN2003 IN3	D4	Motor Phase Control
ULN2003 IN4	D5	Motor Phase Control
10 k Ω Potentiometer	A0	Speed Control Input
Momentary Push-Button	D6	Local Control Input

Table I. ESP32-S3 PIN ASSIGNMENT

C. Stepper Motor:

The 28BYJ-48 5 V geared stepper motor was used to drive the platform.

The motor was selected because:

1. The stepper motor is cost-effective and readily available; hence, it is suitable for low-cost prototype development.
2. It has a small size and therefore can easily fit in any mechanical structure.
3. The motor features an integrated gearing system which provides for increased torque and reduced speed.
4. The stepper motor is capable of moving in a sequence of small steps such that it can rotate the platform slowly.
5. It can be readily interfaced with a ULN2003 driver board for control by the ESP32-S3.

D. ULN2003 Motor Driver:

The ESP32-S3 cannot directly supply the current required by the stepper motor. Hence, a ULN2003 motor-driver board is incorporated between the controller and motor.

The ULN2003 provides the current-switching interface between the low-current ESP32-S3 GPIO signals and the stepper motor windings. Its integrated flyback diodes provide protection against inductive voltage transients.

E. Motor Power Supply:

The stepper motor is powered by an independent 5 V, 2 A supply. This prevents the motor load from being powered through the ESP32-S3. A common ground is maintained between the ESP32-S3 and ULN2003 driver circuit.

F. Potentiometer:

A 10 k Ω potentiometer provides the motor speed-control input. Its wiper signal is connected to analog input A0 of the ESP32-S3.

A decrease in delay between steps results in an increase in rotation speed, and vice versa.

G. Push-Button:

The momentary push-button provides a physical control input to the ESP32-S3. This button can start the rotation cycle programmed into the device depending on how the firmware is set up.

IV. 180° MOTOR CONTROL AND FIRMWARE OPERATION

A. Half-Step Sequence:

STATE	IN1	IN2	IN3	IN4
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1	1	0	0	0
2	1	1	0	0
3	0	1	0	0
4	0	1	1	0
5	0	0	1	0
6	0	0	1	1
7	0	0	0	1
8	1	0	0	1

Table II. Half-Step Excitation Sequence

Advancing sequentially through the table produces incremental motor movement.

B. Forward And Reverse Rotation:

Clockwise and counter-clockwise rotation are obtained by changing the order in which the sequence is executed.

For example:

$$1 - 2 - 3 - 4 - 5 - 6 - 7 - 8$$

produces movement in one direction.

Executing the states in reverse:

$$8 - 7 - 6 - 5 - 4 - 3 - 2 - 1$$

produces movement in the opposite direction.

C. Step-to-Angle Relationship:

The prototype firmware uses a nominal value of:

$$N_{rev} = 4096$$

half-step commands per complete revolution.

The nominal angle associated with one command is:

$$\begin{aligned}\theta_s &= \frac{360^\circ}{4096} \\ &= 0.0879^\circ\end{aligned}$$

For a 180° movement:

$$N_{180} = \frac{180^\circ}{360} \times 4096$$

$$N_{180} = 2048$$

Therefore, approximately 2048 half-step commands are issued for the nominal 180° sweep.

D. Potentiometer-Based Speed Control:

Let T_s represent the delay between two consecutive step commands.

The approximate stepping frequency is:

$$f_s = \frac{1}{T_s}$$

A shorter delay results in higher stepping frequency and a higher delay results in slower movement.

Using the nominal angular displacement per step:

$$\omega = \theta_s f_s$$

where ω represents the approximate platform angular speed.

V. MECHANICAL DESIGN AND TURNTABLE CONSTRUCTION

The lower structure houses the controller, motor driver, stepper motor, wiring, power connections, and mechanical support structure.

The upper platform holds the jewelry and rotates relative to the base.

The space between the upper and the lower structure is approximately 9 cm.

The motor is securely fixed to an internal structure so that the body of the motor remains fixed but only the shaft turns the upper platform.

Careful mechanical alignment is necessary to reduce the following:

1. Platform wobble
2. Shaft misalignment
3. Vibration
4. Uneven rotational loading

The jewelry is placed near the center of the platform to avoid load imbalance.

VI. EXPERIMENTAL TESTING AND RESULTS

A. Testing Procedure:

The prototype was tested through an incremental testing method whereby each subsystem was individually tested before integration. The initial tests included firmware execution of the ESP32-S3, motor-driver operation, bidirectional rotation, input responses, 180° sweep, and Bluetooth Low Energy (BLE) communication. The testing of these features individually facilitated easy detection of faults prior to assembly of the electronic system into the mechanical housing.

While setting up the controller for the first time, there were some problems encountered in communicating over USB/DFU to upload the firmware. These issues were resolved after adjusting the communication configuration of the board.

B. Motor Operation and Direction Control:

The motor-driver interface was tested by executing the programmed half-step sequence on the ESP32-S3 and applying the resulting control signals to the ULN2003 driver. Successful rotation confirmed correct operation of the ESP32-S3–ULN2003–stepper motor control chain.

The motor was tested to operate in both directions. It successfully switched directions upon reversing the excitation sequence. This showed the bidirectional motion needed to move from one end point to the other within the 180° range of motion.

C. User-Input Verification:

Momentary push-button was tested in order to confirm that the physical signals were successfully detected by the controller.

Potentiometer was also tested during the process of motor rotation. Changing positions of the knob led to changes in speed of rotation, which confirmed that the speed control method described in Section IV-D works properly.

D. 180° Sweep Test:

The first functionality check was done using the preset 180° rotation sequence. The controller transmitted the predefined number of commands equal to the necessary number for half-rotation, and the turntable successfully performed the rotation.

The results show that the firmware is able to limit the platform rotation angle to the necessary one instead of full

rotation. Also, the reverse command returned the platform toward its nominal starting position. This result verifies nominal 180° operation; however, precision angular accuracy was not evaluated using an external angular measurement system.

E. Bluetooth Low Energy Test:

The BLE functionality was initialized on the ESP32-S3 and successfully connected to the external device. This demonstrates that wireless communication may be implemented without an additional Bluetooth module.

VII. DISCUSSION AND LIMITATIONS

A. Interpretation of the Experimental Results:

The experimental testing demonstrates the prototype's ability to implement the required functionality of an automated jewelry positioning station.

The successful 180° turn is especially important in this case since the project is focused on a limited angle motion of the mechanism and not on constant rotation.

The ability to vary both the direction and speed of the turntable makes the system suitable for a controlled product-photography operation.

Another feature of the system is that all control and communication functionality has been integrated into the ESP32-S3 module. Thus, the number of external electronic modules is reduced.

The prototype demonstrates the possibility of automating a repetitive element of the product photography process using the small embedded system.

It should be noted that not all elements of photography process have been automated but only the first one.

B. Angular Accuracy:

The functional test does not establish precision angular accuracy because the actual platform orientation is not measured directly.

It is currently impossible to know the actual orientation of the platform through the system. In other words, some discrepancy could arise between the programmed rotation and the actual mechanical rotation.

C. Open-Loop Operation:

The positioning mechanism implemented in the existing prototype utilizes an open-loop technique. The controller estimates platform position from the number of step commands issued to the motor, but receives no independent feedback confirming the actual physical position.

For lightweight loads under ideal operating conditions, this approach is adequate for functional positioning.

This shortcoming needs to be taken into consideration when talking about the positioning abilities of the prototype.

D. Gearbox Backlash:

The internal gearing of the 28BYJ-48 motor may introduce gearbox backlash. This phenomenon will become more obvious when the motor's direction of rotation is reversed as it needs some movement of the motor before the gears mesh again in the opposite direction.

The present design purposely involves back and forth movement, and hence the issue of backlash could have an effect on the repeatability of the return position.

Gearbox backlash was not quantitatively evaluated in the present study.

E. Mechanical Constraints:

The prototype is intended for lightweight jewelry products and a lightweight rotating platform. The selected motor is suitable only for relatively smaller loads.

Greater weight of the platform or placing of the product in an eccentric position would add to the load on the motor and might cause vibrations, rough rotations or skipped steps.

Alignment would also be very important. An eccentricity between the axis of the motor and the rotating platform would lead to wobble.

F. Scope of the Present Prototype:

The present prototype performs automation in positioning of the jewelry alone.

The camera operates independently, and special-purpose high-CRI photography lights have not been used either. This should not be considered in the results of the experiments conducted, since it is not yet an available feature.

In this way, the boundaries of the project will remain clear, as well as the difference between the performed system and its future extensions.

VIII. CONCLUSION AND FUTURE ENHANCEMENTS

A. Conclusion:

The ESP32-S3-based 180° jewelry-positioning turntable was designed and experimentally tested. The experimental testing demonstrated successful bidirectional movement, adjustable speed, local user control, nominal 180° operation, and BLE connectivity.

The prototype provides a practical method for changing jewelry orientation without manually repositioning the product between viewing angles. The mechanical construction is small enough for the main control and driving elements to be hidden under the rotating plate.

Thus, the current work lays the foundation of the positioning platform for further automation of the process of jewelry photography. The next step will involve developing the system for taking photographs and controlling the illumination.

B. Automated Camera Triggering:

The first planned enhancement is automated camera triggering.

Currently, the operator manually triggers the camera after positioning the jewelry on the turntable. In a future implementation, the controller could automatically trigger the camera once the turntable reaches the required angular position.

The planned procedure would involve four steps:

1. Rotate the table to the desired angle
2. Stop the motor
3. Allow a short settling delay for mechanical vibration to decrease
4. Trigger image capture

This sequence could be repeated at predetermined angular positions of 0°, 30°, 60°, 90°, 120°, 150° and 180°. This would enable seven consistent images to be captured with reduced manual intervention.

Also, the automation of the trigger would synchronize the two processes, as pictures would be taken only once the motion stops.

C. High-CRI Lighting:

The second planned enhancement is the integration of high-CRI lighting.

High-CRI lighting has not been utilized or tested in the current prototype but it is proposed for a future implementation.

There are specific problems associated with jewelry photography that relate to reflections and highlights on metallic surfaces. The apparent color of stones, polished metals and reflective surfaces may vary with the illumination conditions.

Thus, diffused high-CRI LED lighting may be employed in the next generation setup to provide better:

1. Color Reproduction
2. Visibility of surface details
3. Consistency across multiple photography sessions

Lighting will be part of the setup but not connected directly to the ESP32-S3. If the ESP32-S3 is used to switch or dim the lighting, an appropriate external driver circuit should be used.

D. Integration of the Future Photography Sequence:

With both enhancements implemented, the turntable, lighting, and camera could operate as a coordinated system.

In order to capture an image for each orientation of jewelry, the turntable would need to move and stop first before any lighting and camera trigger would take place. The controller would then go to the next orientation.

This extension would preserve the present 180° turntable design while expanding its role from automated product positioning to coordinated multi-angle jewelry image acquisition.

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